Topic: Equal real Eigenvalues with multiplicity three

Question: Find the general solution to the system of differential equations.

$$x_1'(t) = x_1$$

$$x_2'(t) = 2x_1 + 2x_2 - x_3$$

$$x_3'(t) = x_2$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t + c_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t$

B $\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t \right)$

C $\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2}{2} e^t + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} te^t + c_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t$

D $\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t \right)$

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Solution: D

We can represent this system in matrix form as

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} \vec{x}$$

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 2 & 2-\lambda & -1 \\ 0 & 1 & -\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} 2-\lambda & -1 \\ 1 & -\lambda \end{vmatrix} - 0\begin{vmatrix} 2 & -1 \\ 0 & -\lambda \end{vmatrix} + 0\begin{vmatrix} 2 & 2-\lambda \\ 0 & 1 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)[(2-\lambda)(-\lambda) - (-1)(1)]$$

$$|A - \lambda I| = (1-\lambda)(-2\lambda + \lambda^2 + 1)$$

$$|A - \lambda I| = -2\lambda + \lambda^2 + 1 + 2\lambda^2 - \lambda^3 - \lambda$$

$$|A - \lambda I| = 1 - 3\lambda + 3\lambda^2 - \lambda^3$$

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Solve the characteristic equation to find the Eigenvalues.

$$1 - 3\lambda + 3\lambda^2 - \lambda^3 = 0$$

$$\lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$$

$$(\lambda - 1)(\lambda - 1)(\lambda - 1) = 0$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we find

$$A - 1I = \begin{bmatrix} 1-1 & 0 & 0 \\ 2 & 2-1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$A - 1I = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn these first two rows into equations, we get

$$k_1 = 0$$

$$k_2 - k_3 = 0$$

So $k_2 = k_3$, and we can choose $k_2 = k_3 = 1$.

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

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Other than the trivial solution $\overrightarrow{k_1} = (0,0,0)$, any other (k_1, k_2, k_3) set we find from $(k_1, k_2, k_3) = (0, k_2, k_3)$ will result in a vector that's linearly dependent with the $\overrightarrow{k_1} = (0,1,1)$ vector we already found. So we can say that the Eigenvalue $\lambda_1 = \lambda_2 = \lambda_3 = 1$ produces only one Eigenvector.

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t$$

Because we only find one Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we have to use $\overrightarrow{k_1} = (0,1,1)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$2p_1 + p_2 - p_3 = 1$$

$$p_2 - p_3 = 1$$

Substituting $p_2 - p_3 = 1$ into the first equation gives

$$2p_1 + 1 = 1$$

$$2p_1 = 0$$

$$p_1 = 0$$

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We can choose any values for p_2 and p_3 that satisfy $p_2 - p_3 = 1$, so we'll pick $(p_2, p_3) = (1,0)$ to get

$$\overrightarrow{p_1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Then our second solution will be

$$\overrightarrow{x_2} = \overrightarrow{k_1} te^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t$$

Now we'll use $\overrightarrow{k_1} = (0,1,1)$ and $\overrightarrow{p_1} = (0,1,0)$ to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{q_1} = \overrightarrow{p_1}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$2q_1 + q_2 - q_3 = 1$$

$$q_2 - q_3 = 0$$

Substituting $q_2 - q_3 = 0$ into the first equation gives

$$2q_1 + 0 = 1$$

$$q_1 = \frac{1}{2}$$

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Plugging this back into the first equation gives

$$2\left(\frac{1}{2}\right) + q_2 - q_3 = 1$$

$$q_2 - q_3 = 0$$

$$q_2 = q_3$$

We can choose any values for q_2 and q_3, so we'll pick $(q_2, q_3) = (0,0)$ to get

$$\overrightarrow{q_1} = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$$

Ideally, we'd like to simplify the solution by eliminating fractions, so we'll multiply through $\overrightarrow{q_1}$ by 2 to rewrite it as

$$\overrightarrow{q_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Then our third solution will be

$$\overrightarrow{x_3} = \overrightarrow{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \overrightarrow{p_1}te^{\lambda_1 t} + \overrightarrow{q_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\frac{t^2}{2}e^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}te^t + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}e^t$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

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$$\overrightarrow{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t \right)$$

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Topic: Equal real Eigenvalues with multiplicity three

Question: Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right) + c_3 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{t^2}{2} e^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right)$$

B $$\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right)$$

C $$\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

D $$\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right) + c_3 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{t^2}{2} e^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^{2t} \right)$$

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Solution: A

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ -1 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda) \begin{vmatrix} 3-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} - 1 \begin{vmatrix} -1 & 0 \\ 1 & 2-\lambda \end{vmatrix} + 0 \begin{vmatrix} -1 & 3-\lambda \\ 1 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)[(3-\lambda)(2-\lambda) - (0)(0)] - [(-1)(2-\lambda) - (1)(0)]$$

$$|A - \lambda I| = (1-\lambda)(3-\lambda)(2-\lambda) + (2-\lambda)$$

$$|A - \lambda I| = (2-\lambda)^3$$

Solve the characteristic equation to find the Eigenvalues.

$$(2-\lambda)^3 = 0$$

$$\lambda = 2, 2, 2$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we find

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$$A - 2I = \begin{bmatrix} 1-2 & 1 & 0 \\ -1 & 3-2 & 0 \\ 1 & 0 & 2-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn these first two rows into equations, we get

$$k_1 = 0$$

$$k_2 = 0$$

So we can choose $k_3 = 1$ to get

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Other than the trivial solution $\overrightarrow{k_1} = (0,0,0)$, any other (k_1, k_2, k_3) set we find from $(k_1, k_2, k_3) = (0,0,k_3)$ will result in a vector that's linearly dependent with the $\overrightarrow{k_1} = (0,0,1)$ vector we already found. So we can say that the Eigenvalue $\lambda_1 = \lambda_2 = \lambda_3 = 2$ produces only one Eigenvector.

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$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

Because we only find one Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we have to use $\overrightarrow{k_1} = (0,0,1)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$-p_1 + p_2 = 0$$

$$p_1 = 1$$

Substituting $p_1 = 1$ into the first equation gives

$$p_2 = 1$$

We can choose any values for p_3, so we'll pick $p_3 = 0$ to get

$$\overrightarrow{p_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Then our second solution will be

$$\overrightarrow{x_2} = \overrightarrow{k_1} t e^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

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$$\overrightarrow{x_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

Now we'll use $\overrightarrow{p_1} = (1,1,0)$ to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{q_1} = \overrightarrow{p_1}$$

$$\begin{bmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$-q_1 + q_2 = 1$$

$$q_1 = 0$$

Substituting $q_1 = 0$ into the first equation gives

$$0 + q_2 = 1$$

$$q_2 = 1$$

We can choose any value for q_3, so we'll pick $q_3 = 0$ to get

$$\overrightarrow{q_1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Then our third solution will be

$$\overrightarrow{x_3} = \overrightarrow{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \overrightarrow{p_1}te^{\lambda_1 t} + \overrightarrow{q_1}e^{\lambda_1 t}$$

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$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{t^2}{2} e^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right) + c_3 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{t^2}{2} e^{2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right)$$

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Topic: Equal real Eigenvalues with multiplicity three

Question: Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} -1 & 5 & 0 \\ 0 & -1 & 0 \\ 0 & 2 & -1 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t}$$

B $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t} + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} te^{-t} \right)$$

C $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{-t} + c_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{t^2}{2} e^{-t}$$

D $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t} + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{-t} \right)$$

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Solution: B

We'll need to find the determinant $|A-\lambda I|$,

$$|A-\lambda I| = \left| \begin{bmatrix} -1 & 5 & 0 \\ 0 & -1 & 0 \\ 0 & 2 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right|$$

$$|A-\lambda I| = \left| \begin{bmatrix} -1 & 5 & 0 \\ 0 & -1 & 0 \\ 0 & 2 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right|$$

$$|A-\lambda I| = \begin{vmatrix} -1-\lambda & 5 & 0 \\ 0 & -1-\lambda & 0 \\ 0 & 2 & -1-\lambda \end{vmatrix}$$

$$|A-\lambda I| = (-1-\lambda)\begin{vmatrix} -1-\lambda & 0 \\ 2 & -1-\lambda \end{vmatrix} - 5\begin{vmatrix} 0 & 0 \\ 0 & -1-\lambda \end{vmatrix} + 0\begin{vmatrix} 0 & -1-\lambda \\ 0 & 2 \end{vmatrix}$$

$$|A-\lambda I| = (-1-\lambda)[(-1-\lambda)(-1-\lambda)-(0)(2)] - 5[(0)(-1-\lambda)-(0)(0)]$$

$$|A-\lambda I| = (-1-\lambda)(-1-\lambda)(-1-\lambda)$$

Solve the characteristic equation to find the Eigenvalues.

$$(-1-\lambda)(-1-\lambda)(-1-\lambda) = 0$$

$$\lambda = -1,\ -1,\ -1$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = -1$, we find

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$$A-(-1)I=\begin{bmatrix}-1-(-1) & 5 & 0\\ 0 & -1-(-1) & 0\\ 0 & 2 & -1-(-1)\end{bmatrix}$$

$$A-(-1)I=\begin{bmatrix}0 & 5 & 0\\ 0 & 0 & 0\\ 0 & 2 & 0\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}0 & 5 & 0\\ 0 & 0 & 0\\ 0 & 2 & 0\end{bmatrix}\rightarrow\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 0\\ 0 & 2 & 0\end{bmatrix}\rightarrow\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

If we turn this first row into an equation, we get

$$k_2=0$$

So we can choose $k_1=1$ and $k_3=0$ to get

$$\vec{k_1}=\begin{bmatrix}1\\0\\0\end{bmatrix}$$

We could also choose $k_1=0$ and $k_3=1$ to get

$$\vec{k_2}=\begin{bmatrix}0\\0\\1\end{bmatrix}$$

So we can say that the Eigenvalue $\lambda_1=\lambda_2=\lambda_3=-1$ produces only two Eigenvectors.

$$\vec{x_1}=\vec{k_1}e^{\lambda_1 t}\qquad\qquad\vec{x_2}=\vec{k_2}e^{\lambda_2 t}$$

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$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t}$$

Because we find two Eigenvectors for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = -1$, we have to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2}$$

$$\begin{bmatrix} 0 & 5 & 0 \\ 0 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ a_2 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$5p_2 = a_1$$

$$2p_2 = a_2$$

We can choose any value for p_2, so we'll pick $p_2 = 1$ with $p_1 = 0$ and $p_3 = 0$.

$$\overrightarrow{p_1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Then our third solution will be

$$\overrightarrow{x_3} = \overrightarrow{p_1}e^{\lambda_1 t} + (a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2})te^{\lambda_1 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t} + \left(5\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) + 2\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) \right) te^{-t}$$

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$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} te^{-t}$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t} + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} te^{-t} \right)$$

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